

Magnetic Field and Magnetic Forces, solved

Step-by-step solutions to every practice problem in Chapter 18. Try each problem on your own before you read its solution.

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P18.1

- **Speed: no.** The magnetic force is always perpendicular to the velocity, so it does no work.
- **Kinetic energy: no,** for the same reason ($\mathbf{W} = 0$, so $\Delta K = 0$).
- **Velocity: yes.** It changes the *direction* of the velocity, bending the path into a circle or helix.

Changing the speed of a charged particle needs an electric field.

P18.2

Electric field lines start and end on electric charges. Magnetic fields are produced by moving charges (currents) and by the intrinsic magnetic moments of particles, never by isolated magnetic “charges”. Every source of magnetic field is effectively a current loop, and the field lines of a current loop close on themselves. Mathematically, $\oint \vec{B} \cdot d\vec{A} = 0$: the net magnetic flux out of any closed surface is zero, because no magnetic monopoles exist inside. If a monopole were ever found, field lines could begin or end on it.

P18.3

Radii. $r = mv/(eB)$, with the same v , B and $|q|$, so $r \propto m$. The proton’s circle is $m_p/m_e = 1836$ times larger.

Periods. $T = 2\pi m/(eB)$. The proton’s period is also 1836 times longer.

Senses of rotation. The charges are opposite, so the forces are opposite, and the two particles circle in **opposite** directions.

P18.4

(a) No. The magnetic force is $q\vec{v} \times \vec{B}$, which is zero when $\vec{v} = 0$. (A stationary charge can still feel an *electric* force, including one from the electric field induced by a *changing* magnetic field, as Chapter 20 shows.)

(b) Yes, if it moves **parallel or antiparallel** to $\vec{B}$, since then $\vec{v} \times \vec{B} = 0$. It also passes undeflected if an electric field exactly cancels the magnetic force, as in a velocity selector.

P18.5

Radius:

$$r = \frac{mv}{qB} = \frac{(6.64 \times 10^{-27})(1.0 \times 10^6)}{(3.20 \times 10^{-19})(0.20)} = 0.10 \text{ m.}$$

Period:

$$T = \frac{2\pi m}{qB} = \frac{2\pi(6.64 \times 10^{-27})}{(3.20 \times 10^{-19})(0.20)} = 6.5 \times 10^{-7} \text{ s.}$$

P18.6

Components: $v_{\perp} = v \sin 30^\circ = 2.5 \times 10^6 \text{ m/s}$ and $v_{\parallel} = v \cos 30^\circ = 4.33 \times 10^6 \text{ m/s}$.

Radius:

$$r = \frac{m_e v_{\perp}}{eB} = \frac{(9.11 \times 10^{-31})(2.5 \times 10^6)}{(1.602 \times 10^{-19})(0.010)} = 1.4 \times 10^{-3} \text{ m} = 1.4 \text{ mm.}$$

Pitch. The period is

$$T = \frac{2\pi m_e}{eB} = 3.57 \times 10^{-9} \text{ s,}$$

so the pitch is

$$p = v_{\parallel} T = (4.33 \times 10^6)(3.57 \times 10^{-9}) = 1.5 \times 10^{-2} \text{ m} = 1.5 \text{ cm}.$$

The helix is long and thin: it advances about 11 times its radius per turn.

P18.7

$$F = ILB \sin \theta = (3.0)(0.50)(0.80) \sin 40^\circ = 0.77 \text{ N}.$$

The force is **perpendicular to both the wire and the field**, in the direction of $\vec{L} \times \vec{B}$ given by the right-hand rule.

P18.8

Force on the rod. The field is vertical and the current horizontal along the rod, so the force is horizontal, along the rails:

$$F = ILB = (4.0)(0.20)(0.50) = 0.40 \text{ N}.$$

$$a = \frac{F}{m} = \frac{0.40}{0.050} = 8.0 \text{ m/s}^2.$$

Distance in 0.50 s: $d = \frac{1}{2}at^2 = \frac{1}{2}(8.0)(0.25) = 1.0 \text{ m}.$

This is a miniature *railgun*. In practice, as the rod speeds up, its motion induces an EMF that opposes the current (Chapter 20), so the acceleration does not stay constant.

P18.9

Magnetic moment: $\mu = NIA = 50(0.50)(0.010) = 0.25 \text{ A m}^2.$

Torque:

$$\tau = \mu B \sin 30^\circ = (0.25)(0.30)(0.50) = 3.8 \times 10^{-2} \text{ N m}.$$

Work to rotate the coil from the aligned position ($\theta = 0$) to 30° :

$$W = \Delta U = (-\mu B \cos 30^\circ) - (-\mu B) = \mu B(1 - \cos 30^\circ) = 0.075(0.134) = 1.0 \times 10^{-2} \text{ J}.$$

P18.10

Radius, from $r = \frac{1}{B} \sqrt{\frac{2mV}{q}}$:

$$r_{235} = \frac{1}{0.25} \sqrt{\frac{2(235)(1.66 \times 10^{-27})(2000)}{1.602 \times 10^{-19}}} = \frac{1}{0.25} \sqrt{9.74 \times 10^{-3}} = 0.395 \text{ m.}$$

$$r_{238} = 0.395 \sqrt{\frac{238}{235}} = 0.395(1.0064) = 0.397 \text{ m.}$$

Separation after a semicircle: $2 \Delta r = 2(0.395 \times 0.0064) = 5.0 \text{ mm.}$

Context. During the Manhattan Project, “calutrons”, which were giant mass spectrometers, separated ^{235}U this way. The separation is small, and the throughput was very low. Today gas centrifuges are used instead.

P18.11

Flow speed. At equilibrium, the electric force from the induced voltage balances the magnetic force on the ions: $q(V/d) = qvB$. So

$$v = \frac{V}{Bd} = \frac{80 \times 10^{-6}}{(0.040)(0.0050)} = 0.40 \text{ m/s.}$$

Volume flow rate:

$$Q = vA = 0.40 \times \pi(2.5 \times 10^{-3})^2 = 7.9 \times 10^{-6} \text{ m}^3/\text{s.}$$

Converting: $7.9 \times 10^{-6} \text{ m}^3/\text{s} \times 10^6 \text{ mL/m}^3 \times 60 \text{ s/min} = 470 \text{ mL/min.}$

The signal is only microvolts. Practical flowmeters use alternating fields and careful amplification to separate it from electrode noise. The great advantage of the method is that it measures flow without disturbing it.

P18.12

(a) Larmor frequencies:

- At 1.5 T: $42.58 \times 1.5 = 63.9 \text{ MHz.}$
- At 3.0 T: $42.58 \times 3.0 = 127.7 \text{ MHz.}$

(b) **Slice difference.** Over 1.0 mm, the gradient changes the field by

$$\Delta B = (10 \times 10^{-3} \text{ T/m})(1.0 \times 10^{-3} \text{ m}) = 1.0 \times 10^{-5} \text{ T}.$$

$$\Delta f = 42.58 \text{ MHz/T} \times 1.0 \times 10^{-5} \text{ T} = 430 \text{ Hz}.$$

(c) **Locating the signal.** With the gradient switched on, the resonant frequency depends on position. A radio pulse with a narrow band of frequencies excites only the protons in one thin slice: *slice selection*. Further gradients, applied along the other directions during excitation and read-out, encode position within the slice in the signal's frequency and phase. A Fourier transform then turns the mixture of frequencies back into a map of where the signal came from. Stronger gradients give finer resolution.

P18.13

(a) **Period.** $T = 2\pi r/v$ with $r = mv/(qB)$ gives

$$T = \frac{2\pi m}{qB},$$

which does not involve v . Faster particles move on larger circles and take exactly the same time per orbit. So an alternating voltage at a **fixed** frequency kicks the particles in step at every crossing of the gap, however fast they become.

(b) **Maximum energy.** At the extraction radius R , $v = qBR/m$, so

$$K = \frac{1}{2}mv^2 = \frac{q^2 B^2 R^2}{2m}.$$

(c) **Deuteron:**

$$f = \frac{eB}{2\pi m} = \frac{(1.602 \times 10^{-19})(1.5)}{2\pi(3.34 \times 10^{-27})} = 1.1 \times 10^7 \text{ Hz} = 11 \text{ MHz}.$$

$$K = \frac{(1.602 \times 10^{-19})^2(1.5)^2(0.60)^2}{2(3.34 \times 10^{-27})} = 3.1 \times 10^{-12} \text{ J} = 19 \text{ MeV}.$$

(d) **High energies.** As v approaches c , the period grows with energy: relativistically, $T = 2\pi\gamma m/(qB)$. The particles fall out of step with the fixed-frequency voltage and stop gaining energy. Solutions:

- the **synchrocyclotron**, which lowers the frequency as the particles speed up;

- the **isochronous cyclotron**, whose field strengthens with radius to keep the period constant (used in proton therapy);
- the **synchrotron**, in which the particles travel on a fixed ring while both the field and the frequency are ramped up.

P18.14

(a) **Derivation.** Let the strip have width w (across the strip, perpendicular to both the current and $\vec{B}$) and thickness t (along $\vec{B}$). In steady state, the Hall field balances the magnetic force on the carriers: $qE_H = qv_d B$. So

$$V_H = E_H w = v_d B w.$$

With $I = nqv_d(wt)$, that is $v_d = I/(nqwt)$:

$$V_H = \frac{IB}{nqt}.$$

(b) **Copper:**

$$V_H = \frac{(10)(1.0)}{(8.5 \times 10^{28})(1.602 \times 10^{-19})(1.0 \times 10^{-4})} = 7.3 \times 10^{-6} \text{ V} = 7.3 \mu\text{V}.$$

(c) **Doped silicon:**

$$V_H = \frac{(1.0 \times 10^{-3})(1.0)}{(1.0 \times 10^{21})(1.602 \times 10^{-19})(1.0 \times 10^{-4})} = 0.062 \text{ V} = 62 \text{ mV}.$$

That is about 10 000 times larger, with 10 000 times *less* current.

Why semiconductors. $V_H \propto 1/n$. Semiconductors have roughly 10^7 times fewer carriers than metals, so each carrier must drift much faster to carry the same current, which produces a much larger Hall voltage. The signal is then easy to measure with simple electronics, and the power consumption is tiny.

P18.15

(a) **The semicircle.** Put the centre at the origin, with the semicircle running from $\theta = 0$ to $\theta = \pi$ in a plane perpendicular to $\vec{B}$. Each element $I d\vec{l}$ is tangent to the arc, so $I d\vec{l} \times \vec{B}$ points radially, with magnitude $IBR d\theta$. By symmetry, the components along the diameter cancel. The

components perpendicular to the diameter add:

$$F = \int_0^\pi IBR \sin \theta d\theta = 2IBR.$$

A straight wire from one end of the semicircle to the other has length $2R$ and feels $I(2R)B$, the same.

(b) **Closed loops.** In a uniform field,

$$\vec{F} = \int I d\vec{l} \times \vec{B} = I \left(\int d\vec{l} \right) \times \vec{B}.$$

The integral of $d\vec{l}$ is just the vector from the start of the wire to its end, so only the end points matter. For a **closed** loop, the start and end coincide, so $\oint d\vec{l} = \vec{0}$ and the net force is zero. A current loop in a uniform field can feel a torque, but no net force. A net force needs a *non-uniform* field.

P18.16

(a) **Energy splitting:**

$$\Delta E = 2\mu B = 2(1.41 \times 10^{-26})(3.0) = 8.5 \times 10^{-26} \text{ J}.$$

$$\frac{\Delta E}{h} = \frac{8.46 \times 10^{-26}}{6.63 \times 10^{-34}} = 1.28 \times 10^8 \text{ Hz} = 128 \text{ MHz}.$$

This matches the Larmor frequency $42.58 \times 3.0 = 127.7$ MHz. The radio photons used in MRI have exactly the energy needed to flip a proton between the two orientations.

(b) **Fractional excess:**

$$\frac{\mu B}{kT} = \frac{(1.41 \times 10^{-26})(3.0)}{(1.38 \times 10^{-23})(310)} = \frac{4.23 \times 10^{-26}}{4.28 \times 10^{-21}} = 9.9 \times 10^{-6}.$$

Only about **10 protons per million** are, on balance, aligned with the field. All the rest cancel each other out.

Consequences. The MRI signal comes from this tiny surplus, so it is intrinsically weak. Doubling the field doubles the surplus *and* raises the signal frequency, which also increases the detected voltage. The signal-to-noise ratio therefore rises roughly in proportion to B , or faster. This is why there is a push towards 3 T and 7 T scanners, and why images take minutes to acquire, since many repeated measurements must be averaged.

P18.17

(a) **Qualitative picture.** The proton starts to accelerate along $\vec{E}$. As soon as it is moving, the magnetic force $q\vec{v} \times \vec{B}$ bends its path sideways. It slows down as it turns back against $\vec{E}$, comes momentarily to rest, and the cycle repeats. The result is a looping path, a **cycloid**, that drifts sideways overall.

(b) **Drift.** Consider a frame moving at $\vec{v}_d = (E/B)\hat{i}$. In that frame, the effective electric field is

$$\vec{E}' = \vec{E} + \vec{v}_d \times \vec{B} = E\hat{j} + \frac{E}{B}\hat{i} \times B\hat{k} = E\hat{j} - E\hat{j} = \vec{0}.$$

There, the proton feels only the magnetic field. It starts with velocity $-(E/B)\hat{i}$ in this frame, so it moves in a circle at constant speed E/B . Back in the lab frame, its velocity is the drift E/B (along $+x$) plus the circular motion of speed E/B . The speed is therefore 0 at the cusps of the cycloid and $2E/B$ at the tops of the loops, and the average motion is the drift $\vec{E} \times \vec{B}/B^2$ along $+x$.

(c) **Drift speed:** $v_d = E/B = 1.0 \times 10^4 / 0.10 = 1.0 \times 10^5$ m/s.

The $\vec{E} \times \vec{B}$ drift does not depend on the particle's charge or mass. It moves whole plasmas, and it matters in fusion devices, in the Earth's magnetosphere, and in magnetron microwave tubes.

P18.18

(a) **Radius.** The proton's speed is

$$v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2(1.602 \times 10^{-13})}{1.673 \times 10^{-27}}} = 1.38 \times 10^7 \text{ m/s.}$$

$$r = \frac{m_p v}{eB} = \frac{(1.673 \times 10^{-27})(1.38 \times 10^7)}{(1.602 \times 10^{-19})(1.1 \times 10^{-6})} = 1.3 \times 10^5 \text{ m} \approx 130 \text{ km.}$$

That is small compared with the scale of the field (about 19 000 km), so the proton stays tied to its field line.

(b) **Period:**

$$T = \frac{2\pi m_p}{eB} = \frac{2\pi(1.673 \times 10^{-27})}{(1.602 \times 10^{-19})(1.1 \times 10^{-6})} = 0.060 \text{ s.}$$

(c) **Bouncing and aurorae.** A particle spiralling along a field line towards the poles moves into a region where the field lines converge and the field grows stronger. As it does, more and more of its

kinetic energy goes into the circular part of its motion, and its speed along the field line falls. At some point the forward motion stops and reverses: the converging field acts as a **magnetic mirror**. The particle therefore bounces between the northern and southern mirror points every few seconds, trapped in the Van Allen belts. Particles whose mirror points lie low enough strike the upper atmosphere near the poles, exciting oxygen and nitrogen atoms, which glow. This happens around the ends of the field lines that connect to the outer magnetosphere, producing the aurora in rings (*auroral ovals*) centred on the magnetic poles.

All worked examples and practice problems in these notes are original. Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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